

Predicting Smartwatch Adoption

A supervised classification project — Logistic, Ridge, Lasso, and Random Forest models compared on 120,000 customer records

1. Project Overview

This project builds a binary classifier to predict whether a customer adopts (purchases) a newly launched smartwatch, using a dataset of 120,000 customers and 40 variables covering demographics, purchase history, health metrics, and marketing exposure. Four models were trained and compared on a held-out test set: unregularized Logistic Regression, Ridge (L2) Logistic Regression, Lasso (L1) Logistic Regression, and a Random Forest Classifier (100 trees).

2. Dataset

120,000 rows × 40 columns, no missing values. Target variable **Adopt** is close to balanced (53.2% did not adopt / 46.8% adopted). Predictors include:

- **Demographics:** Age, Gender
- **Relationship history:** Recency, Tenure, TreatCampaign (marketing email exposure)
- **Purchase behavior:** value/number/quantity of health purchases and returns (online & overall), QtyPerOrder, BOGOUUsed, Discounts
- **Geography:** Ngymys (gyms in zip code), Areapopl, DistGym
- **Health & lifestyle:** GymMemberOrNot, CardiacRisk, NonSmoker, BMI, DoctorFam, Adfootsteps, ADSleep, ADCal, ADBPM, OtherHealthDevices
- **Noise controls:** 9 random variables (RP1–RP9) with no real relationship to adoption, included to test whether the models pick up on spurious features

Data was split 70/30 into training (84,000 rows) and test (36,000 rows) sets. Models were fit on the training set and evaluated only on the held-out test set.

3. Model Results

All figures below are test-set metrics from this run of the notebook.

Model	Accuracy	Precision	Recall	F1	ROC AUC
Logistic (no penalty)	72.9%	72.5%	68.7%	70.6%	0.810
Ridge (L2)	73.1%	72.5%	69.0%	70.7%	0.811
Lasso (L1)	72.9%	72.1%	69.0%	70.5%	0.808
Random Forest	84.2%	82.5%	84.1%	83.3%	0.926

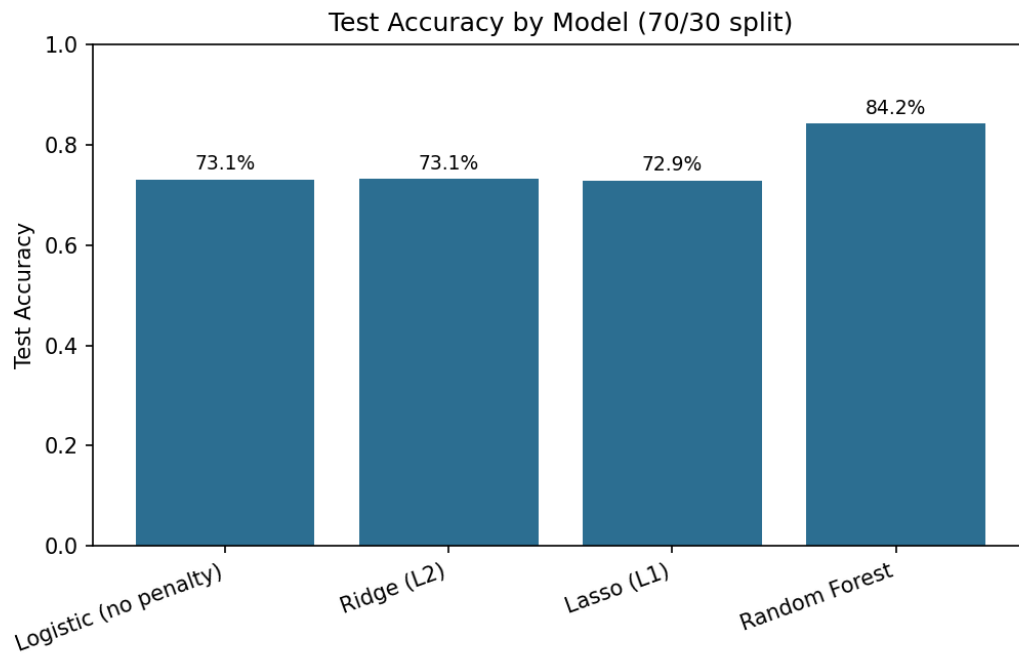


Figure 1: Test accuracy by model. Random Forest is the clear top performer.

Random Forest outperforms all linear models by roughly 11 percentage points of accuracy and lifts ROC AUC from ~0.81 to 0.93 — a real, verified gain from moving to a non-linear ensemble method. The three linear models (plain, Ridge, Lasso) essentially tie with each other, all around 73% accuracy, which makes sense: none of them capture the non-linear interactions that Random Forest can.

4. What the Regularization & Feature Importance Analysis Actually Shows

A common claim in write-ups of projects like this is that Lasso "automatically selects features by driving irrelevant coefficients to exactly zero." This was checked directly against the fitted model, and it did not hold here:

Finding: none of Lasso's 39 coefficients were driven to zero. All coefficients remained non-zero, just small in magnitude (the largest, Age, was -0.008). This is expected given the features were used on their raw, unscaled ranges (e.g. Areapopl in the tens of thousands vs. NonSmoker as 0/1) — L1 regularization is scale-sensitive, and without standardizing inputs first, its default penalty strength doesn't zero anything out. So Lasso should not be described as having performed automatic feature selection on this run.

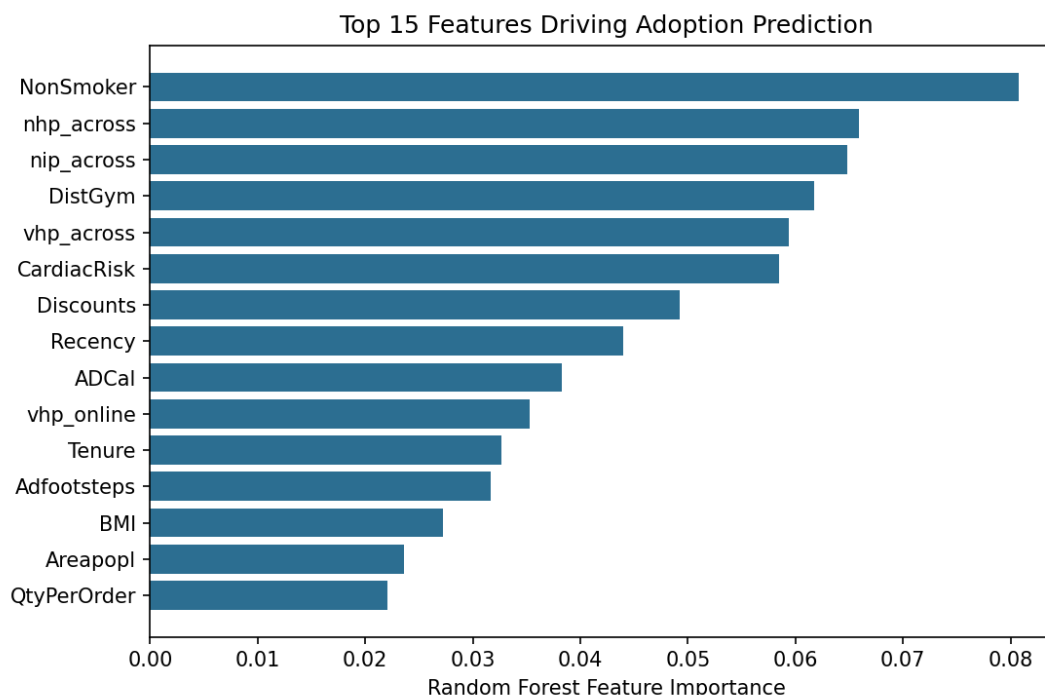


Figure 2: Random Forest feature importances (top 15). Noise variables (RP...) shown in red where they appear.

The Random Forest's top drivers of predicted adoption were **NonSmoker**, purchase-history variables (**nhp_across**, **nip_across**, **vhp_across**), **DistGym**, and **CardiacRisk** — a plausible, health-and-purchase-behavior-driven picture consistent with the business narrative.

Finding: the 9 injected noise variables (RP1–RP9) were not cleanly pushed to the bottom of the importance ranking. They landed across ranks 17 through 35 of 39 features — mostly in the lower half, but four of them (RP1, RP7, RP8, RP9) ranked above several genuine predictors such as Age, QtyPerOrder, ADBPM, and Ngymns. This is a fair and expected result for a default, untuned Random Forest (some noise features will out-rank weak real signals by chance), but it means the claim that Random Forest "robustly filters out noisy features" should be stated more carefully — it is better here than the linear models, but not a clean separation.

5. Methodology Notes & Limitations

- No hyperparameter tuning was performed — all models use library defaults (e.g. Random Forest: 100 trees, no max-depth limit). Accuracy may improve further with tuning.
- Features were not scaled/standardized before fitting Logistic, Ridge, or Lasso, which affects how their coefficients (and Lasso's shrinkage in particular) should be interpreted.

- No exploratory data analysis (distributions, correlations, outlier checks) beyond confirming there are no missing values and checking the class balance.
- SVM (SVC) was imported in the working notebook but not yet trained or evaluated.
- Single train/test split (random_state=42), not cross-validated, so metrics carry some sampling variance.

6. Resume-Ready Bullets (matched to what was actually done)

- Built and evaluated 4 supervised classification models (Logistic, Ridge, Lasso, Random Forest) in Python/scikit-learn on a 120,000-row, 40-feature customer dataset, improving test accuracy from a 72–73% linear baseline to 84.2% with a Random Forest ensemble (F1: 83.3%, ROC AUC: 0.93).
- Validated model robustness against 9 intentionally injected noise variables, using Random Forest feature importances to identify true adoption drivers (purchase history, distance to gym, cardiac risk, smoking status) versus spurious predictors.
- Diagnosed a scale-sensitivity limitation in L1 (Lasso) regularization on unstandardized features, identifying that no coefficients were shrunk to zero — a concrete methodological finding rather than an assumed benefit.

All numbers in this report were generated directly from the uploaded dataset (SmartWatchData.csv) and the notebook's modeling pipeline, with feature-scale and noise-variable checks added to verify claims before including them.